

CASE STUDY / NETWORK SUPPLY • FIBRE SPLICING • RACK INSTALLATION

Plant networking. From fibre to machine.

SML Limited | Panoli Unit-II

Network job for the new P-3 Domino machine

D-TECH's ordered scope brings together Gigabit switching, single-mode optical connectivity, fibre termination and a serviceable rack installation. The drawings explain how the components support the client's machine-network requirement.

2

GIGABIT SWITCHES

2

SFP MODULES

1

6U WALL RACK



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DRAWING-LED PROJECT VIEW

2D connection plan + 3D rack sketch

Illustrative design reconstructed from the ordered components; route, device positions and as-built details require site records.

Client requirement | The network job and ordered scope

The installation service order identifies a network requirement for the new P-3 Domino machine. The job combines the required active devices and fibre termination hardware with splicing, rack installation and cable dressing, giving the client a defined equipment package and installation scope.

Material / component	Qty	Specification in PO 4700000094
D-Link fibre pigtails	12	Pigtail; connector type is not specified on this line.
Duplex optical fibre patch cords	2	PO wording also says “D-LINK 12 PORT”; confirm exact supplied part, connector and length.
D-Link network switches	2	10-port Gigabit description; exact switch model is not stated.
D-Link SFP modules	2	DEM-310GT SM; 1000BASE-LX, single-mode LC.
D-Link wall-mount rack	1	6U, 550 W × 450 D (mm); NWR-6U-5045-GR.
D-Link loaded LIUs	2	12-port LC single-mode termination units.
Cable management panels	2	Rack cable-dressing / patch organisation components.

INSTALLATION ORDER | PO 4700000095

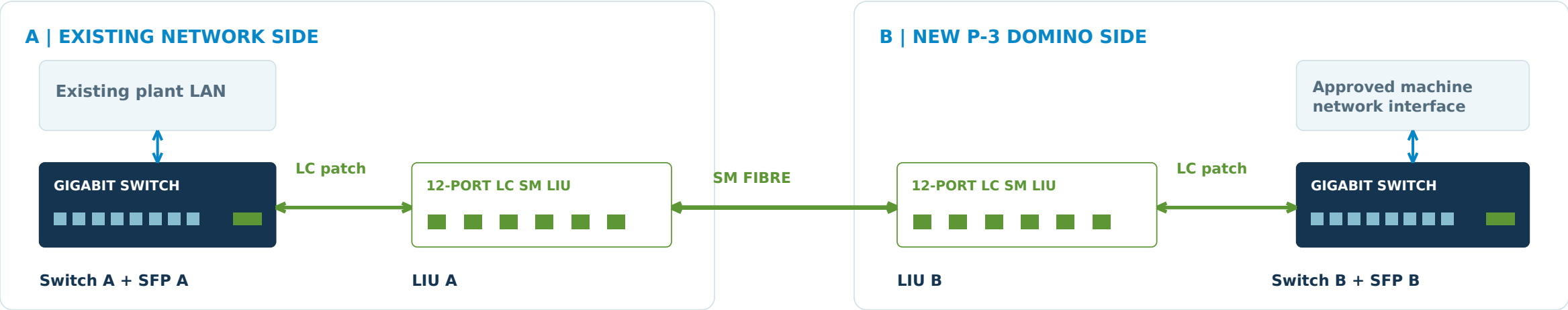
Fibre splicing: 1 AU work unit. Rack installation and dressing: 1 AU work unit. Both services reference Unit-II and the new P-3 Domino machine; the splicing work unit is not a splice-count specification.

Both POs: 3 February 2023 | 03 Connectivity drawing | 04 Rack sketches | 05 Work process & testing | 06 Client & job record

LOGICAL CONNECTION DRAWING / LOCATIONS ARE ILLUSTRATIVE

2D connectivity | A two-ended single-mode fibre link

Two ordered switches and two SFP modules support this two-ended link concept. Confirm the exact switch model’s SFP support, the available single-mode fibre route and the machine vendor’s approved Ethernet interface before applying the connection plan.



1 new 6U rack shown at side B; mounting at side A is site-dependent.

THE OPTICAL PATH

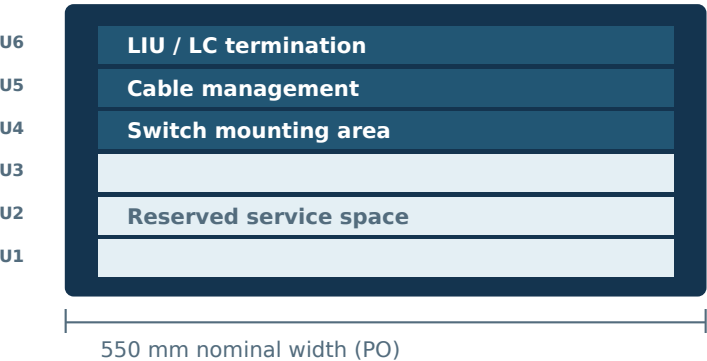
Switch / SFP → duplex LC patch cord → LIU adapter / pigtail → fusion splice → single-mode fibre.

Use a duplex pair with transmit connected to receive at the opposite end. Fibre cable length and core allocation are not stated in the POs.

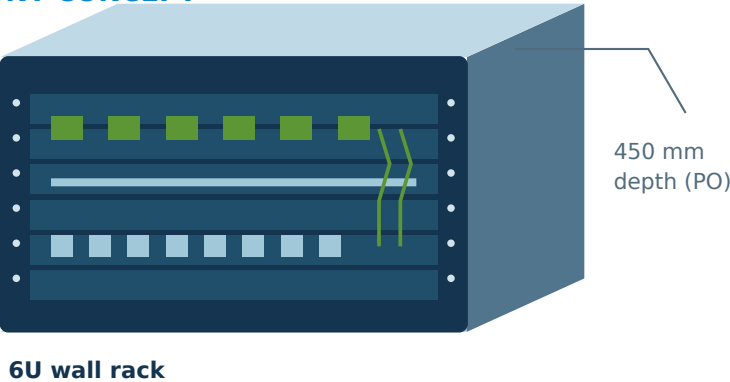
Rack design | A clear, serviceable installation

The rack sketch brings the machine-side fibre termination, patch organisation and switch into an accessible installation. Component positions and mounting arrangements are illustrative; confirm actual LIU size, switch fit, ventilation and cable bend limits on site.

2D | FRONT ARRANGEMENT



3D | WALL-MOUNT CONCEPT



FIBRE TERMINATION

Protect splices in the LIU. Label fibre cores, LC ports and the two link ends; allocate the 12 pigtails against the approved termination plan.

RACK INSTALLATION & DRESSING

Fix to a suitable wall, secure the equipment and dress patch leads through the manager. Confirm power provision, grounding and access with SML.

Illustrative rack-unit positions; the drawing is not a dimensioned fabrication or as-built drawing. The second LIU / switch / manager belongs to the other link end.

Work process | Install, test and hand over

D-TECH’s installation scope is fibre splicing and rack installation / dressing. The workflow below shows how to turn that scope into a documented network job, with SML confirming the route, access, machine interface and final acceptance.



Acceptance area	What to verify	Handover evidence
Physical installation	Rack fixing, safe cable entry, equipment fit, clear labels and dressed leads.	Rack and termination photographs; port / asset list.
Optical termination	Core continuity and Tx / Rx polarity; splice / connector loss and received power within the approved optical budget.	Fibre test report; core and LIU port map.
Network connection	Compatible SFPs; expected negotiated link; machine access through the SML-approved network interface.	Link status and agreed connectivity test record.
Client acceptance	Resolve observations; confirm documentation and owner / support handover.	Client acknowledgement and final job sheet.

The PO requires a test report with every supply. Checks shown here are an acceptance plan; measured test results and a completion certificate were not supplied.

Client & job record | SML Limited, Panoli



SML Limited
Formerly Sulphur Mills Limited
Transforming Agriculture

Job record	Reference
CLIENT SITE	Plot 230/231/232, Panoli, Taluka Ankleshwar, Bharuch, Gujarat 394116. Plant code 1020 / Unit-II.
MATERIAL ORDER	PO 4700000094, dated 3 February 2023. Seven material lines.
INSTALLATION ORDER	PO 4700000095, dated 3 February 2023. Fibre splicing; rack installation and dressing.
SOURCE BASIS	Supplied POs establish the ordered scope. Drawings reconstruct an illustrative design; actual route, switch model and acceptance results require site records.



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Network design, supply, installation & support

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Company image: SML official website | Technical reference: D-Link DEM-310GT | Scope: supplied purchase orders